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New paleomagnetic and paleointensity data from Pliocene lava flows from the Lesser Caucasus

Manuel Calvo-Rathert [a,*], María Felicidad Bógalo [a], Avto Gogichaishvili [a,b], Jemal Sologashvili [c], Goga Vashakidze [d]

[a] Departamento de Física, EPS, Universidad de Burgos, c/Francisco de Vitoria, s/n, 09006 Burgos, Spain
[b] Laboratorio Interinstitucional de Magnetismo Natural, UNAM, Campus Morelia 58098, Mexico
[c] Caucasus International University, Chargali St. 63, 380192 Tbilisi, Georgia
[d] Alexandre Janelidze Institute of Geology – I. Javakhishvili Tbilisi State University, Tbilisi, Georgia

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ABSTRACT

A paleomagnetic, rock-magnetic and paleointensity study has been carried out on 14 basaltic lava flows from two Pliocene (K–Ar age between 3.09 ± 0.10 Ma and 4.00 ± 0.15 Ma) sequences (Apnia and Korxi) from the eastern Djhavakheti Highland in southern Georgia (Caucasus).

Measurement of strong-field magnetisation versus temperature curves yielded three types of thermomagnetic curves: (i) Reversible curves with magnetite as only remanence carrier (type H); (ii) irreversible curves with magnetite as only carrier of remanence (type H^-) and (iii) irreversible curves showing a low Curie-temperature phase and magnetite (type L). Analysis of hysteresis curves showed that samples were characterised by a mixture of single-domain and multi-domain grains.

Paleomagnetic experiments allowed determining characteristic components for all flows and normal polarities (6 flows), reversed polarities (7 flows) and intermediate polarities (1 flow) were observed.. Paleomagnetic poles were calculated using only those sites unequivocally showing normal or reversed polarities. The paleomagnetic pole obtained from flows of both combined sequences (latitude $\lambda = 77.9°$N, longitude $\phi = 152.1°$E, $n = 13$, $A_{95} = 11.8°$, $k = 13.4$) showed a good agreement with the 5 Ma window of the European synthetic apparent polar wander path of Besse and Courtillot (2002). The paleomagnetic direction of the combined Apnia-Korxi flows agrees well with the expected one, showing no significant tectonic rotation. The latter cannot be however, completely excluded in the Korxi section. In that section, analysis of the angular dispersion of virtual geomagnetic poles yields a much higher value than expected.

Paleointensity experiments using the Coe method were performed on 31 specimens from 10 flows. After application of specific selection criteria, 19 samples from 8 flows were observed to provide successful determinations, with mean flow values showing a wide scatter. If only flows with more than one successful paleointensity determination are taken into account, virtual dipole moments (VDMs) vary between 3.5×10^{22} A m^2 and 8.3×10^{22} A m^2. In intermediate polarity site AP2 no weak transitional paleostrength values were observed.

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1. Introduction

Direction and strength of the Earth's magnetic field vary with time and knowledge of the characteristics of the ancient geomagnetic field can provide important information in order to better understand and constrain the processes related to the evolution of the Earth's deep interior, like the generation of the geomagnetic field or the occurrence of polarity reversals. Volcanic rocks allow a reliable and instantaneous record of the Earth's magnetic field by means of the acquisition of thermoremanent magnetisation (TRM) and are able to supply absolute paleointensity data. The magnetisation record provided by volcanic rocks is tied to volcanic eruptions and therefore discontinuous, but because volcanic rocks are in principle able to provide a real image of the Earth's magnetic field, the study of lava flows emitted in a relatively short period of time can be of major interest.

In many paleomagnetic studies, only the directional information provided by declination and inclination of the remanence vector is required. The need for a better knowledge of the variations of the Earth's magnetic field, however, also demands the determination of the strength of the paleofield vector. Unfortunately, the number of reliable paleointensity data available is still limited. The paleointensity database PINT03 (Perrin and Schnepp, 2004)

* Corresponding author. Tel.: +34 947259064.
E-mail address: mcalvo@ubu.es (M. Calvo-Rathert).

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shows that available data are still scarce and unevenly distributed: Approximately a thousand data per Myr for the first million years, 48 determinations per Myr in the 1–20 Ma interval, about 3 determinations per Myr for the 20–400 Ma interval and much less than 0.04 determination per Myr for older units. The paleointensity databases PINT06 (Tauxe and Yamazaki, 2007) and PINT2010 (Biggin et al., 2010), in which some more recent studies (until end of 2010) have been included, show a similar picture.

This is related to the fact that paleointensity determinations are experimentally much more difficult than estimations of the direction of the paleofield vector, and the failure rate of these experiments is often high. In addition, the dispersion observed in paleointensity results is much larger than in directional results, often due to the fact that incorrect determinations are considered to reflect a correct paleointensity value (e.g., Calvo et al., 2002). The paucity of reliable palaeointensity data is directly related to the difficulty of obtaining reliable absolute palaeointensity determinations. This is so because of different reasons: (i) The primary remanence of a rock must be a thermoremanent magnetisation (TRM) in order to be suitable for palaeointensity studies; (ii) Rock samples employed for palaeointensity determinations must obey the Thellier laws of reciprocity, independence and additivity of partial thermoremanent magnetisation (pTRM) acquired in non-overlapping temperature intervals (Thellier and Thellier, 1959). This behaviour is only verified for true SD particles, while multidomain (MD) grains do not obey these laws. Pseudo single-domain (PSD) grains carrying a TRM may approximately satisfy the requirements of the Thellier method for the smallest particles, but not for the larger ones (Shaskanov and Metallova, 1972; Levi, 1977; Bol'shakov and Shcherbakova, 1979; Worm et al., 1988); (iii) During heating, irreversible chemical/mineralogical or physical (Kosterov and Prévot, 1998) changes can affect the magnetic phases, which results in erroneous paleointensity estimates.

Due to its geological and geophysical interest, the Caucasus region has already been the subject of a number of paleomagnetic studies. However, reliable paleomagnetic data from the Caucasus are still sparse if compared to other regions belonging to the Alpine fold belt (Bazhenov and Burtman, 2002), as many of those studies date back several decades and the methodology employed often does not fulfill the minimum reliability and quality criteria required for present-day paleomagnetic results, including, for instance, results based on non-demagnetised samples. In addition, most of those results have only been published in data bases (e.g., Khramov, 1984) and in Russian. Nevertheless, during the last two decades several paleomagnetic studies dealing with different topics like archeomagnetism, paleointensity or tectonics have been published which provide reliable paleomagnetic results from the Caucasus (Bazhenov et al., 1996; Camps et al., 1996; Goguitchaichvili et al., 1997, 2000, 2001, 2009; Goguitchaichvili and Parès, 2000; Bazhenov and Burtman, 2002; Calvo-Rathert et al., 2008, 2011; Shcherbakova et al., 2009), but the amount of recent and trustworthy paleomagnetic data from this region is still limited.

As will be outlined in the *Geologic setting and sampling section* below, the Caucasus is a region of great tectonic complexity and interest due to its position between the converging Eurasian and Arabian plates (e.g., Adamia et al., 2011). Paleomagnetic data are an excellent means to obtain information about possible tectonic rotations experienced by the analysed units of an area in the geological past. On the other hand, lava flows can provide a real image of the strength of the Earth's magnetic field in the time of their formation. Paleointensity data available so far are still relatively scarce and show an uneven geographical distribution. For these reasons, paleomagnetic and paleointensity studies carried out on volcanic rocks from the Caucasus can be of considerable paleomagnetic interest. In this study we present new rock magnetic, paleomagnetic and paleointensity results from the Caucasus area, obtained on samples from 14 lava flows belonging to two volcanic flow sequences, Apnia and Korxi, which are located in the eastern Djavakheti Highland in southern Georgia (Fig. 1).

2. Geologic setting and sampling

The structure and geological evolution of the Caucasus are mainly determined by its location between the still converging Eurasian and Arabian plates within a wide zone of continental collision (e.g., Adamia et al., 2011). Relative motion between both plates has an approximate S–N direction and according to geodetic data, the convergence rate amounts to 20–30 mm yr^{-1} (e.g., Adamia et al., 2011) although a GPS-based global plate motion model (Sella et al., 2002) gives lower estimates of this convergence by approximately 12 mm yr^{-1}. Initial collision between Arabia and Eurasia took place in the Oligocene to middle Miocene and fold–thrust belts were formed at the Greater and Lesser Caucasus during syn-collisional (Oligocene–middle Miocene) and post-collisional (late Miocene–Quaternary) stages (e.g., Adamia et al., 2008).

The region is characterised by the complexity of its active tectonics, showing both N–S compressive (E–W thrusts and folds) and E–W extensional (submeridian normal faults and dykes) structures (Rebaï et al., 1993). Three main directions of active faults compatible with the nearly N–S compression produced by the convergence of the Arabian and Eurasian plates can be distinguished in Georgia (Adamia et al., 2008). The first group of structures is characterised by a WNW–ESE or W–E strike and represented by compressional structures like reverse faults, thrusts and nappes. Two other groups of structures are characterised by either a NE–SW or a NW–SE strike. They are mainly extensional structures but have a considerable strike-slip component (Adamia et al., 2011). The subaerial volcanism of the Djavakheti Highland and some other areas of Georgia is most likely related to processes of W–E extension causing emergence of the system of sub-meridian faults that served as conduits for the rising magma melts (Adamia et al., 2008).

In fact, one of the main characteristics of the Caucasus region is its important and continuous volcanic activity, at least from the Jurassic until present (e.g., Rebaï et al., 1993). The Neogene–Quaternary magmatism in the Caucasus, which is associated to the collision between the Eurasian and Arabian plates (Koronovskii and Demina, 1999) started in middle Miocene and lasted until Holocene, the last volcanic eruptions taking place in historic time (e.g., Aydar et al., 2003). Milanovskii and Koronovskii (1973) distinguished three stages in the evolution of late post-collisional magmatism in the Caucasus: (1) late Miocene to early Pliocene, (2) middle Pliocene to Pleistocene and (3) Quaternary.

The present study was carried out on rocks from the Djavakheti Highland, one of the largest neovolcanic areas of the Lesser Caucasus (Fig. 1). It represents a young tectonic unit formed during the Neogene–Quaternary and is one of the most seismic active regions of the Caucasus (Kachakhidze et al., 2003). The basement of the Djvakheti Highland is composed of Cretaceous and Paleogene volcanogenetic sedimentary rocks, which are exposed in many erosion windows. Throughout a greater part of the region, basement rocks are overlain by the Neogene–Quaternary sequence of volcanic rocks, which can have a thickness of up to hundreds of metres. Volcanic rocks most widespread in the Djavakheti Highland erupted during the middle Pliocene to Pleistocene stage of the late Cenozoic magmatism, and its composition varied from sub-alkaline basalts of mantle origin to rhyolites of predominantly crustal genesis (Lebedev et al., 2008). This subaerial volcanism was most likely related to processes of W–E extension causing emergence of the system of submeridional faults that served as conduits for the rising magma melts (Adamia et al., 2008).

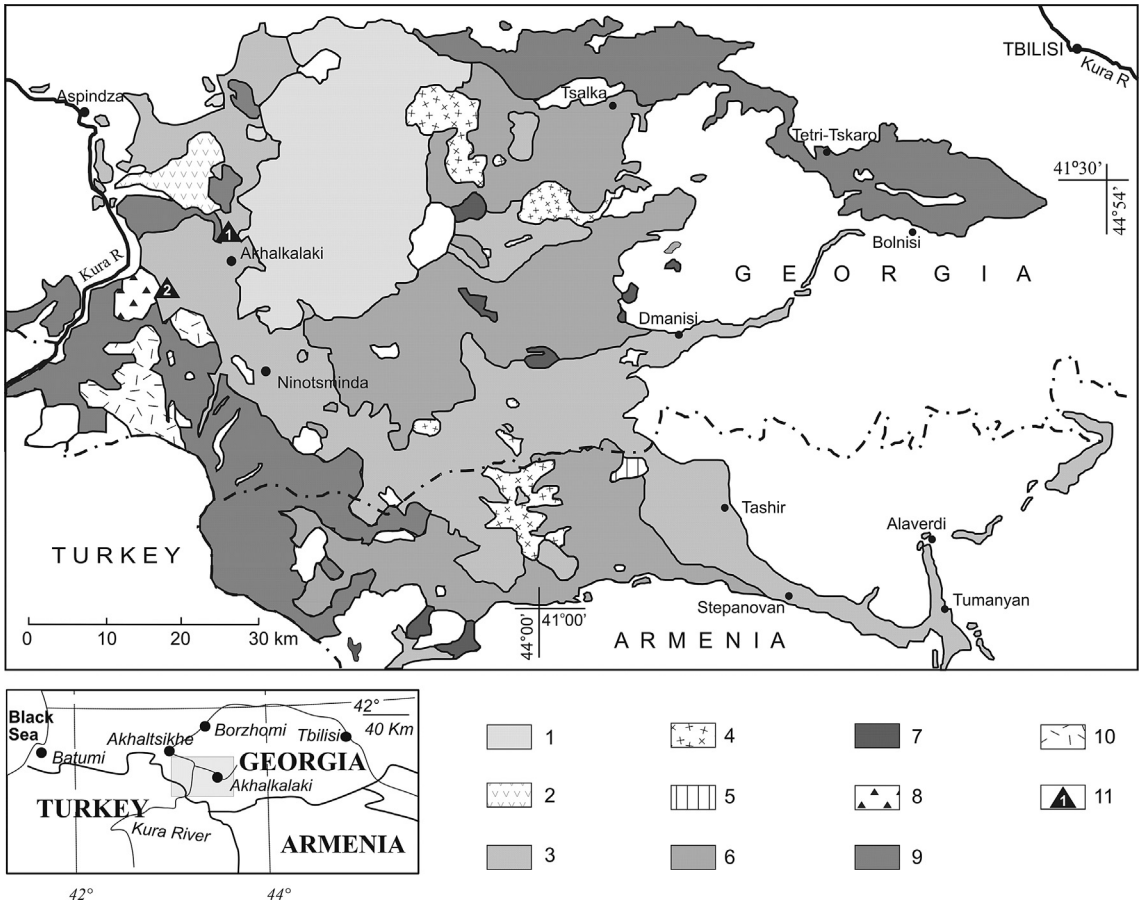


Fig. 1. Schematic geological map (modified after Lebedev et al., 2008) of the Pliocene magmatism in the Djavakheti Highland (Southern Georgia) showing lava flow sequences sampled in the present study. 1: Quaternary volcanics of the Samsari ridge (800–0 ka); 2: Basic lavas (1.75–1.55 Ma); 3: Basic lavas (2.15–1.95 Ma). 4: Dacitic lavas of the Djavakheti ridge (2.25 Ma); 5: Hyalodacite (2.5 Ma); 6: Basic lavas (2.65–2.45 Ma); 7: Rhyolites and dacites of the Chikiani, Agvorik and Busistsikhe volcanoes (2.85–2.6 Ma); 8: Dacites (3.15–3.11 Ma) of the Kumurdo lava flow (a) and Amiranisgora volcano (b); 9: Basic lavas (3.22–3.04 Ma); 10: Basic lavas (3.75–3.55 Ma); 11: Sampled lava flow sequences, (1) Korxi, (2) Apnia.

Some previous paleomagnetic studies carried out on different Pliocene and Quaternary lava flow sequences in the Djavakheti Highland were mainly directed towards paleointensity determinations (Camps et al., 1996; Goguitchaichvili et al., 2009; Calvo-Rathert et al., 2011). Directional results from those studies yielded no rotations of the paleomagnetic vector in the analysed volcanic sequences. No rotations were also detected in a paleomagnetic study carried out by Goguitchaichvili et al. (1997) in the Tchunchka section in that region. Two other previous studies in the Djavakheti area deal with paleomagnetic results obtained on a large number of volcanic flow sequences (Goguitchaichvili et al., 2000, 2001). In the former study, a slightly eastwardly deviated palaeodeclination was obtained for the Djavakheti region while in the latter this appears to be nonrotated. In both cases, however, several individual units display palaeodeclinations clearly diverging from the northern directions. This rotations, however, might be non-significant, as in many of the studied flows only very few samples (1–3) were analysed. The available results for the Korxi section point to a small eastward rotation while the Apnia data display an almost 30° difference in paleodeclination between flows with normal- or reverse-polarity magnetisation (Goguitchaichvili et al., 2001).

The present study is a reconnaissance paleomagnetic, rock-magnetic and paleointensity investigation aimed towards obtaining information to contribute to the description of the variations of the intensity and direction of Earth's magnetic field. In addition, it intends to supply new paleomagnetic data from the Caucasus and specifically from the Djavakheti Highland which might be of interest to obtain more information about the recent tectonics in this region. It was carried out on rocks from two basaltic lava flow sequences of Pliocene age from the eastern Djhavakheti Highland in southern Georgia, the Apnia and Korxi sections.

The Apnia section (41°21′40″N, 43°16′02″E, Fig. 1) outcrops along a gravel road descending from the Akhalkalaki Plateau to the medieval cave monastery of Vardzia and the exposed part of the sequence consists of 20 basaltic lava flows. A K–Ar geochronological study was carried out by Lebedev et al. (2008) on this section and in the uppermost horizon of the sequence they determined an age of 3.09 ± 0.10 Ma. Immediately below, in what the authors call "the second lava horizon of the section" a K–Ar age of 4.20 ± 0.30 Ma was obtained, which they considered an overestimation due to ^{40}Ar excess in the rocks because of incomplete degassing of the magma during quick chilling and crystallization under water. Downward in the section, K–Ar ages of 3.28 ± 0.10 Ma, 3.75 ± 0.25 Ma and 3.70 ± 0.20 Ma were determined and in the lowermost exposed horizon of the sequence a 4.00 ± 0.15 Ma was obtained. This latter result was also interpreted as an overestimated age due to the same reasons as in the second horizon (Lebedev et al., 2008). The Apnia sequence has also been the subject of a previous paleomagnetic study. Sologashvili (1986) obtained normal polarity magnetisations in its upper part and reversed polarity magnetisations in its lower part. However, only very few samples were analysed in each flow and most of them were not fully demagnetised.

The Korxi section (41°27′31″N, 43°27′55″E, Fig. 1) is located at the confluence of the Korxisckali and Paravani rivers and consists of a sequence of massive basaltic lava flows of varying thickness between approximately 10 cm and 10 m. No radiometric data are available for this section A previous paleomagnetic study carried out by Sologashvili (1986) yields both normal and reverse polarity magnetisations, but only very few samples were analysed in each flow and most of them were not fully demagnetised.

The flows of the Apnia and Korxi sequences are formed of microcrystalline rocks pink–brown in colour and porphyritic in texture. The contents of phenocrystals do not exceed 10% and they are represented by idiomorphic, and less frequently xenomorphic crystals of plagioclase (andesine) 0.3–0.7 mm in size and by single grains of hornblende entirely replaced by opacite. Plagioclase crystals are indistinctly zoned. The trachytic groundmass is composed of subparallel plagioclase laths and elongated magnetite crystals, which completely substitute hornblende (Dzhigauri, 1991). Comparison of the specific petrological characteristics of both sequences allowed Lebedev et al. (2004) to estimate the age of the Korxi section to be similar to that of the lower lying Apnia flows which yield ages of 3.70 ± 0.20 and 3.75 ± 0.25 Ma.

Samples used for the present study were taken in 1984–1986 sample collection campaigns. Oriented blocks were sampled from 10 flows in the Apnia section and four flows in the Korxi section and prismatic specimens were cut in the laboratory from the blocks. Orientation relative to the sun was performed by means of a theodolite, and no bedding correction was necessary. Due to the long time elapsed since sampling had been carried out and the inadequate description of the 1984–1986 sampling, it was not possible to find out which specific lava flows from each sequence had been sampled. However, all four sampled Korxi flows correspond to the four lowermost flows of that section. It must be also mentioned at this place that both sequences have a significantly larger number of flows than those sampled for the present study.

3. Rock-magnetic measurements

Rock-magnetic experiments were undertaken to find out the carriers of remanent magnetisation, to determine their thermal stability and grain size and as a pre-selection criterion to choose suitable sites and temperature intervals for paleointensity determinations. The experiments included the measurement of strong-field magnetisation versus temperature (M_S–T), hysteresis parameters and isothermal remanent magnetisation (IRM) acquisition curves. All measurements were performed with a Variable Field Translation Balance (VFTB) in the University of Burgos (Spain). For these measurements, a single whole-rock powdered sample was taken from each flow (although two samples per site were taken for Korxi flows KX3 and KX5) and subjected to the following measurement sequence: (i) IRM acquisition, (ii) backfield, (iii) hysteresis curve, and (iv) M_S–T curve. After having subjected the samples to heating up to approximately 700 °C to record M_S–T curves, hysteresis and backfield curves as well as M_S–T curves were measured again on the heated specimens. This procedure allowed determining hysteresis parameters after thermal treatment in order to compare them with those of untreated samples. In addition, thermomagnetic curves obtained from heated samples were useful to better analyse the magnetic mineralogy resulting from the previous heating procedure.

For the measurement of M_S–T curves, samples were heated in air up to approximately 700 °C and cooled down to room temperature. Curie points were determined using the two-tangent method (Grommé et al., 1969). After the first heating, three types of thermomagnetic behaviour could be distinguished: Type H curves (Fig. 2a) are characterised by a simple behaviour, with low-Ti titanomagnetite as the only carrier of remanence (Table 1) and a high degree of reversibility. This kind of curves was observed in eight cases (seven sites). However, if the difference between initial magnetisation before heating had started and final magnetisation after cooling had been completed was more than 20%, we considered those thermomagnetic curves not to belong to type H.

Type H$^-$ curves (Fig. 2b) were observed in three cases. They are similar to type H samples, but irreversible, as cooling curves display a much weaker magnetisation than heating curves. In one case (flow KX2, Table 1), however, low-Ti titanomagnetite was not the only carrier of remanence and another weak phase with higher Curie temperature was observed in both the heating (T_C = 618 °C) and the cooling curves (T_C = 611 °C). This phase could be also observed in both the heating and the cooling curves obtained from the already heated sample from flow KX2, which showed a reversible type-H thermally stabilised behaviour. The high Curie-temperature might be ascribed to the presence of low-Ti titanohematite in this sample, but the effect of the relatively high fraction of an antiferromagnetic phase suggested by the thermomagnetic curves of this sample should also be noticed in the values of its hysteresis parameters and the shape of its IRM-acquisition curve. All these parameters, however, display no significant differences with other samples from this study. On the other hand, the high Curie-temperature might be attributed to the presence of partially-oxidised magnetite in which the surface layer is maghemitised. The Curie temperature of this phase reflects the degree of oxidation and falls somewhere between magnetite and maghemite Curie-temperatures (e.g., Liu et al., 2001; Gehring et al., 2009). The magnetisation loss mainly observed between the heating and cooling branches of the first thermomagnetic curve can be attributed to the thermally unstable magnetic component, maghemite, inverting to hematite, in accordance with the slight drop in magnetisation and increase in coercivity observed when hysteresis parameters before and after heating are compared.

Type L samples (Fig. 2c) were observed in five cases (four sites). Two phases can be distinguished in the heating curve: A high Curie-temperature phase corresponding to low Ti-titanomagnetite and a low Curie-temperature phase, with Curie temperatures between 100 and 300 °C (Table 1). In the cooling curves, only a single phase (low-Ti titanomagnetite) appears in three cases, while in two other samples also the low Curie-temperature phase can still be recognised. In these latter cases (flows KX5 and AP8), the low-temperature phase is even weakly displayed in the cooling curve of the second thermomagnetic curve determined on the already heated samples.

Most thermomagnetic curves obtained from heated samples are reversible curves with low-Ti titanomagnetite as the only carrier of remanence (Table 1) and a high degree of reversibility, and thus belong to type H. The only exceptions are the already mentioned cases in type-L flows KX5 and AP8 and type H$^-$ flow KX2. In this latter case (named type H*) a fully reversible thermomagnetic curve was observed on the heated sample, but besides low-Ti titanomagnetite also a higher Curie-temperature phase could be recognised in both the heating and the cooling curves.

Hysteresis parameters were obtained from hysteresis and backfield curves and the RockMagAnalyzer 1.0 software (Leonhardt, 2006) was used to analyse the measured curves. As suggested by the Day-plot shown in Fig. 3a, the domain structure of all except one of the studied samples plots in the PSD (pseudo-single-domain) area (Day et al., 1977). This behaviour might be also explained by a mixture of single-domain (SD) and multi-domain (MD) particles (Dunlop, 2002), the proximity to the saturation remanence to saturation ratio (M_{RS}/M_S) value of 0.02 (PSD–MD region boundary; Dunlop, 2002) shown in the figure indicating an increasing amount of the MD component. If these data from the

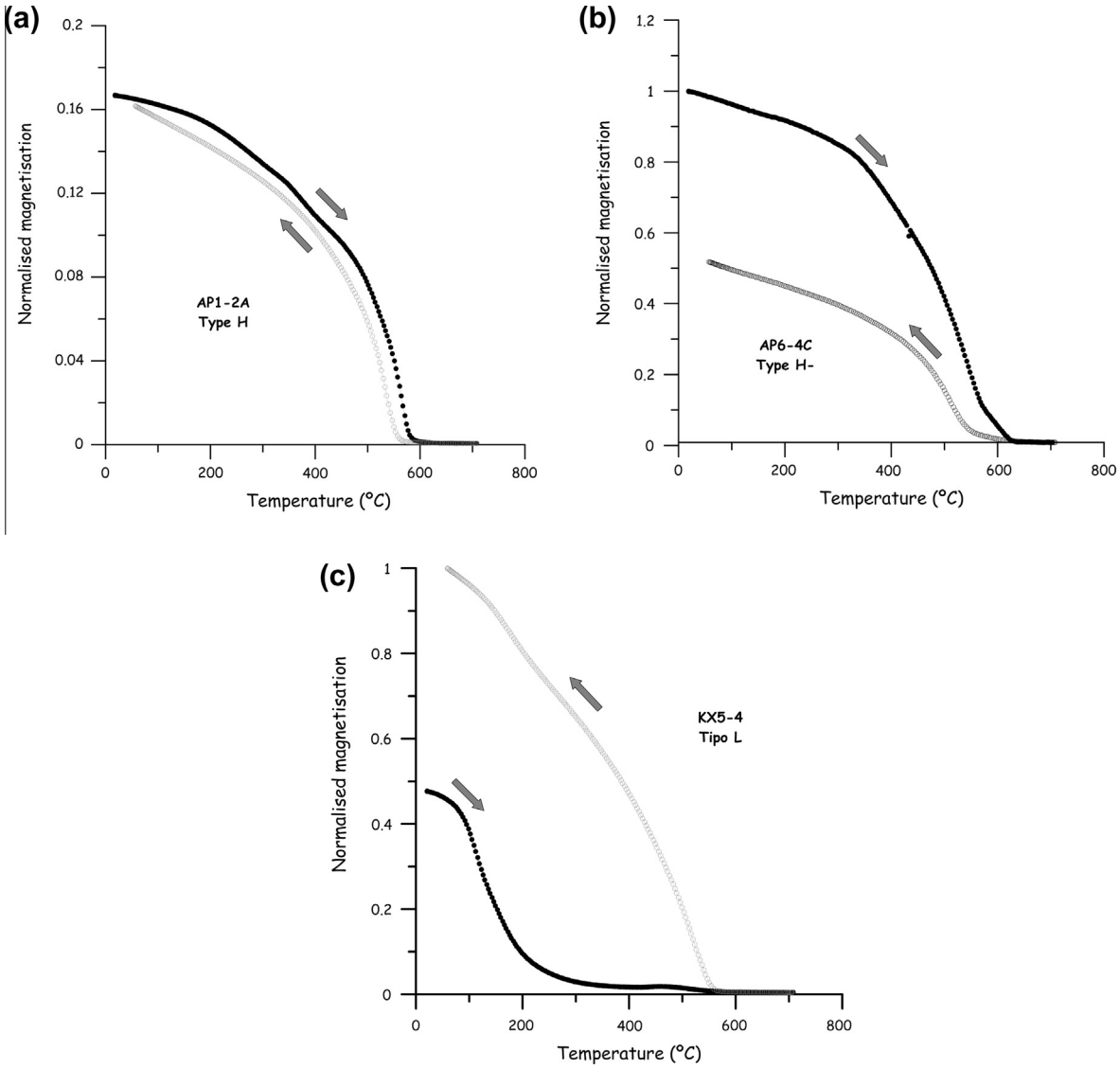


Fig. 2. Normalised strong field magnetisation-versus-temperature curves from lava flow samples from the Apnia and Korxi sequences. (a) Type H curve. Sample AP1-2A (Apnia). (b) Type H$^-$ curve. Sample AP6-4C (Apnia). (c) Type L curve. Sample KX5-4 (Korxi).

present study are compared with theoretical Day plot curves calculated for magnetite (Dunlop, 2002), their relative amount of MD particles in the mixture would vary between approximately 20% and 80%. The only sample showing a SD domain state belongs to flow KX5-6.

Though useful, information provided by Day-plots can be often ambiguous. High M_{RS}/M_S ratios are indicative of the presence of SD or PSD grains as remanence carriers. Low M_{RS}/M_S ratios, however, can be observed both if large MD particles or small superparamagnetic (SP) particles are abundant. Shape parameter σ_{HYS} and coercivity ratio B_{RH}/B_{CR} (Fabian, 2003) may provide some additional information about the domain state of the studied samples. Graphically B_{RH} is the positive field value where the difference between upper and lower hysteresis branches has decreased from $2M_{RS}$ to M_{RS} at $B = 0$. High B_{RH}/B_{CR} ratios indicate large particles, while B_{RH}/B_{CR} ratios below 1 are observed in natural ensembles containing SP particles (Fabian, 2003). Shape anomalies of hysteresis loops are often interpreted as being indicative of mixtures of fractions with highly contrasting coercivities (e.g., Roberts et al., 1995; Muttoni, 1995; Tauxe et al., 1996) originating from mixed assemblages of multiple magnetic components with different mineralogy or grain size. The shape parameter σ_{HYS} gives a quantitative measure related to the shape of the hysteresis loop, with $\sigma_{HYS} > 0$ for wasp-waisted loops and $\sigma_{HYS} < 0$ for pot-bellied loops. This parameter is relatively independent of grain size within the SD–MD region, so that variations in σ_{HYS} are indicative of the presence of SP grains or other mineral fractions (Fabian, 2003).

Shape parameter σ_{HYS} was observed to be negative in all cases, indicating pot-bellied hysteresis curves (Fig. 3b). Fig. 4 shows a plot of the latter parameter versus B_{RH}/B_{CR} ratio. The variation of σ_{HYS} of most samples, with the exception of KX5-4 and KX6-4 (both from flow KX5) is relatively low, varying between -0.62 and -1.05. Thus the larger differences observed in the B_{RH}/B_{CR} ratio among the analysed samples can be mainly ascribed to their SD–MD trend. In both samples belonging to flow KX5-6, however, B_{RH}/B_{CR} is either smaller than 1 or near that value, and σ_{HYS} has somewhat smaller absolute values than those of the remaining samples, and may therefore signal the presence of SP particles.

Hysteresis and backfield curves were measured again after having subjected samples to heating up to 700 °C to record M_S–T curves, so that hysteresis parameters after thermal treatment could be obtained and compared with those of untreated samples. Hysteresis curves after heating still have a "pot-bellied" shape, as indicated by shape parameter σ_{HYS}. The relation between

Table 1
Rock-magnetic results. *Sample*: Sample name. *Flow*: Flow name. *Before H.*: Hysteresis parameters obtained before heating the samples. 1st *Heating*: Data from thermomagnetic curves after the first heating. *After H.*: Hysteresis parameters obtained after heating the samples. 2nd *Heating*: Data from thermomagnetic curves after the second heating. M_{RS}/M_S: Remanent saturation to saturation magnetisation ratio. B_{CR}/B_C: Coercivity of remanence to coercivity ratio. *CT*1(*H*): 1st Curie temperature obtained from the heating curve of magnetisation-versus-temperature experiments. *CT*2(*H*): 2nd Curie temperature obtained from the heating curve of magnetisation-versus-temperature experiments. *CT*1(*C*): 1st Curie temperature obtained from the cooling curve of magnetisation-versus-temperature experiments. *Type*: Thermomagnetic curve type (see text).

Sample	Flow	Before H.		1st Heating					After H.		2nd Heating				
		Mrs/Ms	Bcr/Bc	CT1(H)	CT2(H)	CT1(C)	CT2(C)	Type	Mrs/Ms	Bcr/Bc	CT1(H)	CT2(H)	CT1(C)	CT2(C)	Type
AP1-2A	AP1	0.27	1.68	579		554		H	0.32	1.65	561		553		H
AP2-1B	AP2	0.18	2.26	575		557		H	0.21	2.17	556		545		H
AP3-1B	AP3	0.16	1.77	536	178	554		L	0.37	1.62	542		528		H
AP4-2A	AP4	0.39	1.58	580		575		H	0.38	1.49	579		579		H
AP5-2A	AP5	0.16	2.67	575		550		H	0.19	2.87	544		544		H
AP6-4C	AP6	0.31	2.19	585		547		H⁻	0.22	3.67	558		548		H
AP7-2C	AP7	0.24	2.05	571		561		H	0.24	2.17	558		558		H
AP8-1C	AP8	0.12	2.12	551	118	536	141	L	0.35	2.09	544	156	540	107	L
AP9-1D	AP9	0.27	1.78	582		550		H	0.30	1.77	554		548		H
AP10-3B	AP10	0.37	1.64	582		561		H⁻	0.41	1.62	566		561		H
KX1-1	KX1	0.14	2.10	543	278	535		L	0.21	2.38	541		541		H
KX2-1	KX2	0.24	1.93	551	618	533	611	H⁻	0.27	1.90	546	617	535	610	H*
KX3-4	KX3-4	0.17	2.24	570		554		H	0.19	2.32	558		551		H
KX4-3	KX3-4	0.20	1.69	556		556		H	0.24	1.79	556		563		H
KX5-4	KX5-6	0.22	2.95	588	200	555	236	L	0.40	1.47	554	208	542	178	H(L)
KX6-4	KX5-6	0.64	1.26	555	268	555		L	0.42	1.54	551		551		H

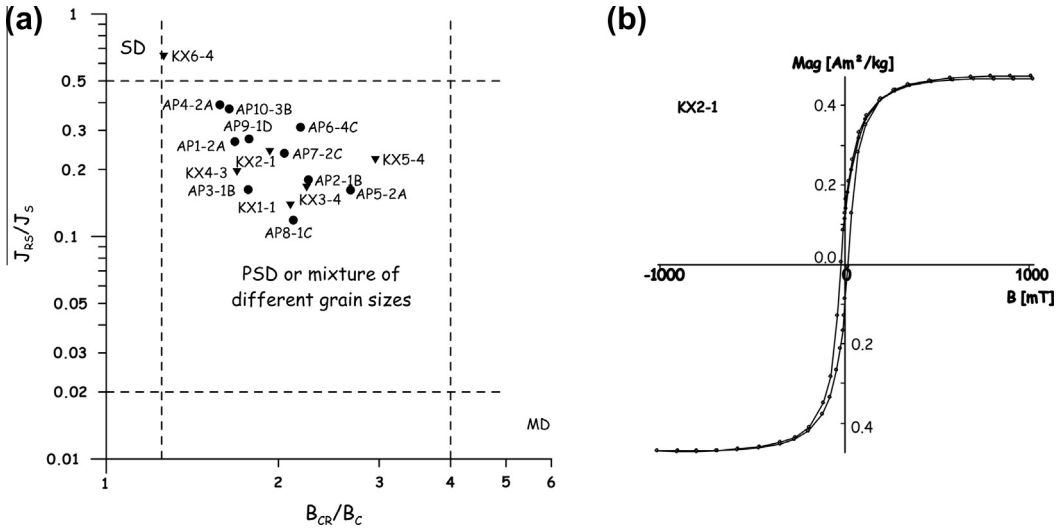


Fig. 3. (a) Bi-logarithmic Day-plot (Day et al., 1977) modified after Dunlop (2002). J_{RS}/J_S: Saturation remanence to saturation magnetisation. B_{CR}/B_C: Coercivity of remanence to coercivity. Samples from each of both sections are shown with different symbols: *Korxi* (inverted black triangles); *Apnia* (black circles). (b) Hysteresis curve of sample KX2-1 (Korxi).

hysteresis-parameter ratios M_{RS}/M_S and B_{CR}/B_C before and after heating is shown in Fig. 5a. In most samples, a moderate enhancement of the M_{RS}/M_S ratio can be observed, but in some type-L samples this increase is very strong. In most of these cases the B_{CR}/B_C ratio only shows small changes and most samples trend towards a more "SD-like" behaviour after heating. Two samples (KX6-4 and AP6-4C), however, show a clear decrease of the M_{RS}/M_S ratio and an increase of the B_{CR}/B_C ratio. This means that changes in domain structure have occurred, even if no major mineralogical changes have happened, as can be recognised in the highlighted type H strong-field thermomagnetic curves. The data shown in Fig. 5b show that in most cases heating enhanced coercivity, probably reflecting a decrease in effective grain size. Saturation remanence did not change in most cases (Fig. 5c).

Isothermal remanent magnetisation (IRM) acquisition curves were recorded in a maximum applied field of approximately 1 T. Low-coercivity phases are the main carriers of remanence, and at an applied field of 0.2 T, 85–97% of SIRM (saturation isothermal remanent magnetisation) has already been acquired (Table 1).

4. Paleomagnetic measurements

Paleomagnetic measurements were performed at the paleomagnetic laboratory of the University of Burgos. Remanent magnetisation was measured with a 2G cryogenic magnetometer, thermal demagnetisation was performed with a Schonstedt TSD-1 furnace and AF demagnetisation with a degausser included in the 2G cryogenic magnetometer equipment. Directions of remanent magnetisation components were determined in all cases by means of principal component analysis (Kirschvink, 1980). In most of the studied 14 sites, six to twelve samples were used for paleomagnetic analysis, but in sites AP9, KX1 and KX2 only three samples were available in each case.

As shown in Table 2, directions of natural remanent magnetisation (NRM) were characterised by very low scatter except in sites AP3 and AP8. The most suitable demagnetisation technique was chosen after subjecting in each site a pilot sample to thermal demagnetisation and another pilot sample to AF demagnetisation. In most cases a simple paleomagnetic behaviour could be

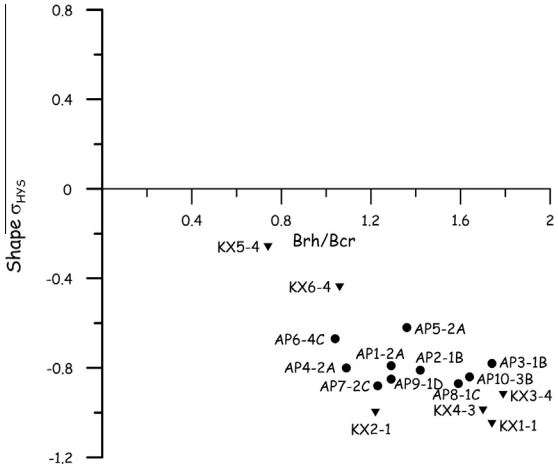


Fig. 4. Plot of shape parameter σ_{HYS} versus B_{RH}/B_{CR} ratio. B_{RH}: coercivity ratio (see text), B_{CR}: coercivity of remanence).

observed, with samples displaying a single main paleomagnetic component (Fig. 6a and b). Nevertheless, in several cases an initial weak overprint could also be distinguished, sometimes up to temperatures about 350 °C. The highest unblocking temperatures detected normally lie in the range of the Curie temperature of magnetite. In a few cases, however, a low fraction of total magnetisation is erased at unblocking temperatures clearly above 585 °C, pointing to the presence of hematite or maghemite. This high-temperature remanence fraction is nevertheless not related to any different paleomagnetic component. In site AP8, five of eleven samples display a similar simple behaviour, with a viscous approximately present-day direction component at low unblocking temperatures together with a single normal-polarity main paleomagnetic component. Both are slightly more overlapped than in the remaining sites. However, in six samples from site AP8, a high-temperature component is observed between approximately 540 °C and the Curie-temperature of magnetite (Fig. 6c). This component is clearly different from the main paleomagnetic component isolated in site AP8 (Table 2), although due to its relatively weak intensity, paleomagnetic directions determined are in most

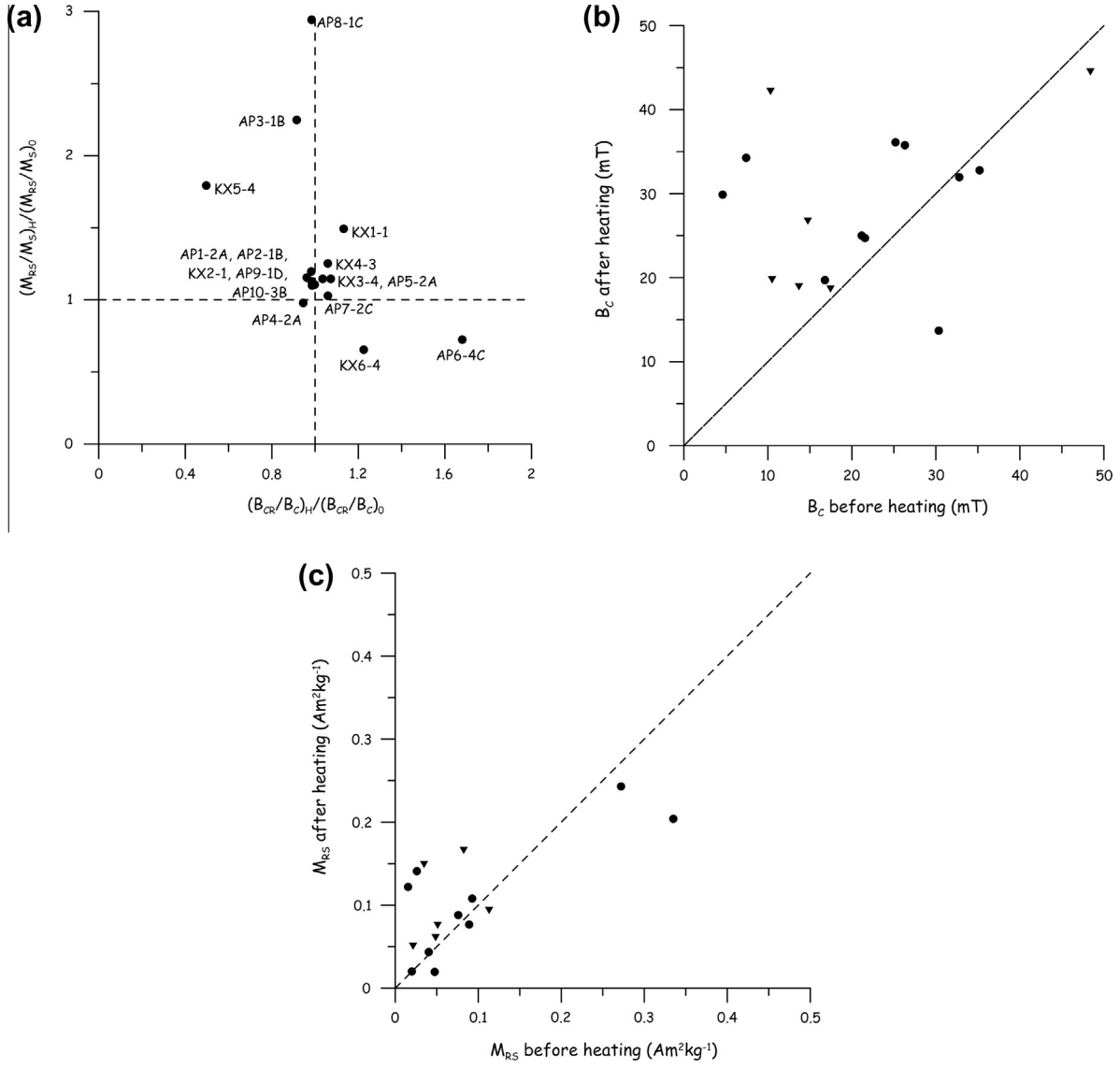


Fig. 5. (a) Relation between hysteresis-parameter ratios M_{RS}/M_S (saturation remanence to saturation magnetisation) and B_{CR}/B_C (coercivity of remanence to coercivity) after and before heating. Subscripts are used to indicate ratios after heating (T) and before heating (0). (b) Relation between coercivity after and before heating. (c) Relation between saturation remanence after and before heating. Symbols as in 3a.

Table 2
Paleomagnetic results. *Flow*: Flow name. *NRM*: Natural remanent magnetisation. *ChRM*: Characteristic remanent magnetisation. *VGP*: Virtual geomagnetic poles. *N*: Number of specimens; in site mean calculations *N* is the number of flows. (*n* + *p*): Number of directly determined directions (*n*) and planes (*p*) used for determination of flow mean. *Dec*: Declination. *Inc*: Inclination. α_{95} (A_{95}): Radius of 95% confidence cone. *k*: Precision parameter. *Lon*: Longitude of site. *Lat*: Latitude of site. Mean ChRM and paleomagnetic poles (calculated using only flows with normal or reversed polarities) are given for the Apnia, Korxi and both combined sequences. Mean directions and statistical parameters of the Apnia-sequence, the Korxi-sequence and the combined flows from both sequences (KX + AP) are indicated with bold characters.

Flow	NRM					ChRM					VGP	
	N	*Dec*	*Inc*	α_{95}	*k*	*N* (*n* + *p*)	*Dec*	*Inc*	α_{95}	*k*	Lon (°E)	Lat (°N)
AP1	7	189.0	−51.7	2.1	823.3	7	190.0	−56.9	2.7	518.9	337.4	−81.3
AP2	12	22.4	−56.2	1.5	820.5	12	21.2	−57.6	1.4	906.2	206.8	8.0
AP3(A)	6	3.2	54.9	17.8	15.1	4 (3 + 1)	326.2	49.4	7.8	155.2	302.0	60.6
(B)						5(0 + 5)	8.9	19.0	28.4	16.2		
AP4	7	214.5	−50.8	1.3	2091.2	7	214.3	−52.1	1.5	1560.6	320.1	−61.5
AP5	8	194.8	−44.6	1.6	1182.2	8	194.3	−46.2	1.5	1288.1	358.9	−71.8
AP6	12	231.3	−46.0	1.8	608.5	12	230.7	−46.3	0.8	3089.8	315.5	−46.7
AP7	6	351.8	60.3	2.5	700.6	6	351.1	61.5	1.9	1290.6	326.6	83.3
AP8 (A)	11	345.1	46.1	7.6	37.3	11	336.3	39.2	3.6	163.6	276.7	62.3
(B)						6	25.0	28.4	25.5	7.9		
AP9	3	170.0	−58.0	3.7	1087.2	3	184.0	−61.3	2.3	2883.7	294.5	−86.9
AP10	12	201.5	−53.0	0.8	2830.5	12	202.4	−55.2	0.8	3140.3	324.3	−71.6
MEAN						**9**	**7.6**	**55.0**	**11.9**	**19.6**	**173.1**	**81.8**
APNIA											A_{95} **= 15.6°**	k **= 11.9**
KX1	3	42.9	67.0	5.6	485.5	3	50.5	69.3	6.5	358.4	97.0	54.7
KX2	3	18.9	46.2	1.6	5890.9	3	18.4	46.5	1.4	7543.7	169.9	69.7
KX3-4	6	22.6	48.7	2.2	968.8	6	21.9	48.4	2.0	1095.7	160.7	68.6
KX5-6	6	211.9	−51.6	3.8	306.1	6	211.4	−55.0	4.0	283.7	316.8	−64.9
MEAN						**4**	**27.6**	**55.3**	**14.4**	**41.5**	**135.5**	**67.5**
KORXI											A_{95} **= 18.3°**	k **= 26.9**
MEAN						**13**	**13.8**	**55.5**	**8.9**	**22.5**	**152.1**	**77.9**
KX + AP											A_{95} **= 11.8°**	k **= 13.4**

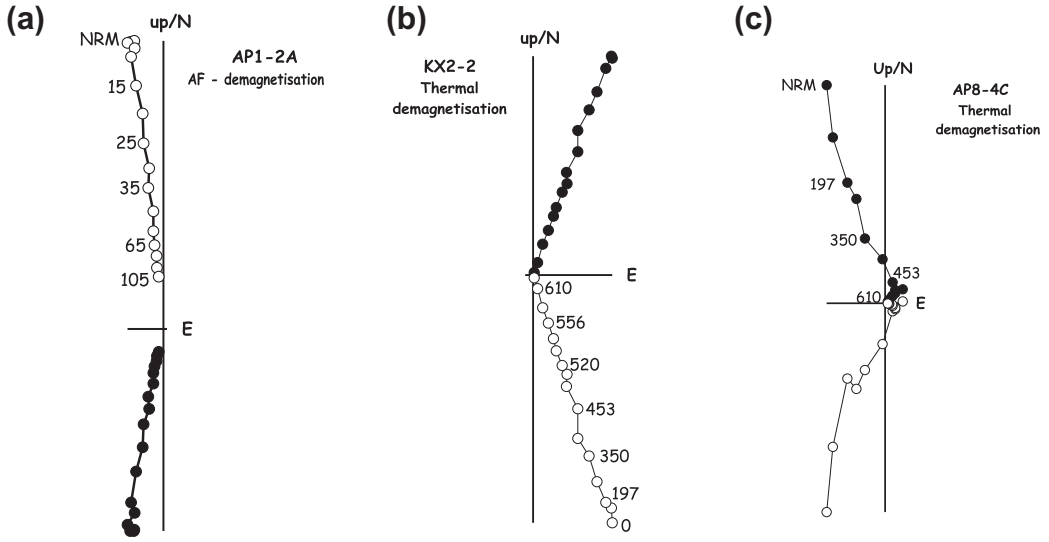


Fig. 6. Orthogonal demagnetisation vector plots. (a) AF-demagnetisation of AP1-2A (Apnia). (b) Thermal demagnetisation of KX2-2 (Korxi). (c) Thermal-demagnetisation of AP8-4C (Apnia). Solid symbols are for the horizontal projection and open symbols for the vertical projection.

cases of not very good quality and the scatter observed among determinations from different specimens is rather high. Samples from site AP3, on the other hand, showed a clearly more complex behaviour than those of the remaining sites. Differing demagnetisation behaviours and different (in some cases apparent) paleomagnetic components can be observed on samples from this site. This may be related to the fact that in several cases overlapping unblocking temperature or coercivity spectra hinder direct determination of a paleomagnetic component in certain sectors of the demagnetisation path or produce relatively straight intervals in demagnetisation plots which might be erroneously taken for real paleomagnetic components. A paleomagnetic component with unblocking temperatures between approximately 100 and 500 °C was determined combining three directly determined directions with a remagnetisation circle (McFadden and McElhinny, 1988) (component A in Table 2) and analysis of remagnetisation circles from five specimens provided another paleomagnetic component with a significantly different direction (component B in Table 2). It should be taken into account that thermomagnetic curves measured for sites AP3 and AP8 belong to type L (Table 1) and are characterised by high thermal instability (Fig. 5a).

Table 2 displays mean ChRM directions determined for each site. In sites AP3 and AP8 we have taken component A as ChRM, as component B is in both cases poorly defined as indicated by high α_{95} and low *k* values. VGPs obtained in the present study are also shown in Table 2 and it can be recognised that normal polarities were obtained in six cases (sites AP3, AP7, AP8, KX1, KX2 and KX3-4) and unequivocally reversed polarities in six cases (sites AP1, AP4, AP5, AP9, AP10 and K5-6). Doubts arise, however, in site AP6. While site AP2, which has a VGP latitude $\lambda = 8.0°$ can be

clearly considered to carry a transitional direction, site AP6, in which a VGP latitude $\lambda = -46.7°$ was observed, could either record an intermediate or a reverse polarity direction. Often the latitude of virtual geomagnetic poles (VGPs) is compared with the pole of the Earth's rotation axis and a fixed cut-off angle of 40° or 45° is adopted to discriminate between secular variation and intermediate polarity directions. A method proposed by Vandamme (1994), however, allows determining an optimum cut-off angle to estimate the most precise value of the angular standard deviation (ASD) of paleosecular variation. If this calculation is performed for the VGPs of the Apnia section considering their angular deviation with respect to their mean, an optimum cutoff angle of 49° is obtained and only site AP2 appears to carry a transitional direction. A similar result is obtained for the combined Apnia-Korxi VGP dataset, Site AP6 thus should be considered to carry a reverse-polarity direction.

A question to be posed is if ChRM directions listed in Table 2 are primary paleodirections. Reversed and intermediate polarity directions determined in eight sites indicate that ChRM has been successfully isolated from normal present-day secondary directions. If both sets of six normal and seven reversed polarity directions are compared, they pass the reversal test of McFadden and McElhinny (1990) with a type-C classification, which points to a primary origin of the determined ChRM directions, although it does not guarantee the latter. In addition, analysis of hysteresis parameters (Figs. 3a and 4) has shown, as discussed above, that in most sites a mixture of SD and MD grains is observed, so that at least part of the remanent magnetisation should be considered to be stable. In fact, in flow KX5-6 SD behaviour is observed. In site AP8 two paleomagnetic components can be distinguished (Table 2). The fact that in this site several specimens only carry component A up to unblocking temperatures reaching the Curie-temperature of magnetite without showing any presence of component B seems to support that component A is the primary remanence of these samples. Nevertheless, it cannot be completely ruled out that component A found in site AP3 is not the primary remanence of those samples, although in one of six studied specimens it is the only paleomagnetic component found.

5. Paleointensity experiments and results

Paleointensity determinations were carried out using a Thellier type double heating method (Thellier and Thellier, 1959) as modified by Coe, 1967) at the LIMNA paleomagnetic laboratory of the *Universidad Nacional Autónoma de México* in Morelia (Mexico). The experiments were performed on non-oriented samples under air using a ASC TD-48 paleointensity oven. 31 samples from 10 sites were selected for these determinations after analysing paleomagnetic data and rejecting sites AP3, AP8 and KX1 (type L samples). No more specimens were available to carry out paleointensity determinations on site AP4. The experiment was carried out in 12 or 13 temperature steps between room temperature and 595 °C or 629 °C, and the laboratory field intensity was set to 40 μT and held at a precision better than 0.15 μT. PTRM-checks were performed at certain temperature steps. Temperature reproducibility between two heatings to the same temperature was generally within 3 °C up to 450 °C and 2 °C above. . Remanence was measured with a JR6 spinner magnetometer.

Paleointensity determinations were carried out with the aid of the ThellierTool4.0 software (Leonhardt et al., 2004a). Whether a paleointensity determination is considered to be reliable depends on a set of chosen criteria regarding the quality of the experiment, the occurrence of alteration and the presence of MD related remanent magnetisation. Although several authors have proposed statistical parameters and reliability criteria for paleointensity determinations (e.g., Selkin and Tauxe, 2000; Kissel and Laj, 2004), in contrast to conventional paleomagnetic analyses, no specific set of criteria and parameters is generally used. Nevertheless, selection criteria do not differ markedly among different paleointensity studies. In the present study, in order to be considered acceptable, paleointensity determinations had to satisfy all of the following requirements:

(a) On the NRM–pTRM diagram the number of aligned points $N \geqslant 4$, without considering data suspected to correspond to viscous magnetisation (VRM) acquired in situ. The selected temperature segment for paleointensity analyses must match the temperature range which carries the characteristic remanence of the studied sample.
(b) NRM fraction factor f (Coe et al., 1978) $\geqslant 0.35$. The fraction factor f (as well as the slope of the best fit line the Arai plot, gap factor g and the quality factor q) was determined for a chosen segment of the Arai diagram. The fraction factor f is referred to the so-called "true NRM", which is the intersection between linear fit and y-axis Leonhardt et al. (2004a).
(c) The ratio β (standard error/absolute slope of the slope of the best fit line on the Arai plot) $\leqslant 0.1$.
(d) The quality factor q (Coe et al., 1978) $\geqslant 2$, with ($q = f \cdot g/\beta$).
(e) DRAT (difference ratio, Selkin and Tauxe, 2000) of PTRM-checks < 7°. The DRAT, is the difference between the pTRM check and original TRM value at a given temperature divided by the length of the segment of the Arai plot used for determining the paleointensity.
(f) Paleointensity results obtained from NRM–pTRM diagrams must not have a clearly concave up shape, as in such cases remanence is probably related to the presence of multidomain grains (Levi, 1977). In the present study this was assessed visually. In addition, also the values of the normalised pTRM-tail $d(t^*)$ as proposed by Leonhardt et al. (2004b) were taken into account. The authors consider that critical values for distinguishing between SD, PSD and MD ranges would approximately correspond to $d(t^*) \leqslant 2.5\%$ for SD grains, $2.5\% \leqslant d(t^*) \leqslant 7\%$ for PSD grains and $d(t^*) > 7\%$ for MD grains. Thellier simulations performed by Leonhardt et al. (2004b) yield deviations of <10% from the correct field intensity as long as $d(t^*) < 5\%$. We have considered that a minimum acceptance criterion should both include the absence of concave-up shape of Arai plots together with a $d(t^*)$ value $\leqslant 7\%$.
(g) Directions of NRM end-points at each step obtained from paleointensity experiments must draw a straight line pointing to the origin in the interval chosen for paleointensity determination. They should define a straight line with mean angular deviation (MAD) smaller than 7°.

Though not exactly the same, this array of selection criteria is similar to the PICRIT-03 set proposed by Kissel and Laj (2004). If these criteria are applied, 19 samples out of 31 studied (61%) provide successful paleointensity determinations (Fig. 7a and Table 3).

Table 3 summarises results of successful paleointensity determinations. Several samples yielding accepted results seemed to be affected by VRM overprints in the lower temperature range, but this happened always below the range chosen for paleointensity determination. Most excluded samples do not fulfil criterion *f*), regarding the presence of MD grains, as shown by the concave-up shape of Arai diagrams (Fig. 7b). No paleointensity determinations were observed to yield normalised pTRM-tail $d(t^*)$ values above 7%, but paleointensity experiments on samples AP6-4D, AP10-3D and KX4-2 were considered unsuccessful determinations due to MD behaviour despite $d(t^*)$ values lying clearly below 7%. Three out of five paleointensity determinations from site

AP2 were discarded due to their low quality factor $q < 2$ and/or high scatter of selected points in the Arai plot. Only sample AP5-4A was rejected because of the occurrence of magneto-mineralogical alterations.

6. Discussion of results

Rock-magnetic, paleomagnetic and paleointensity measurements have been carried out on 14 basaltic lava flows of Pliocene age from the Apnia and Korxi sections, which are located in the

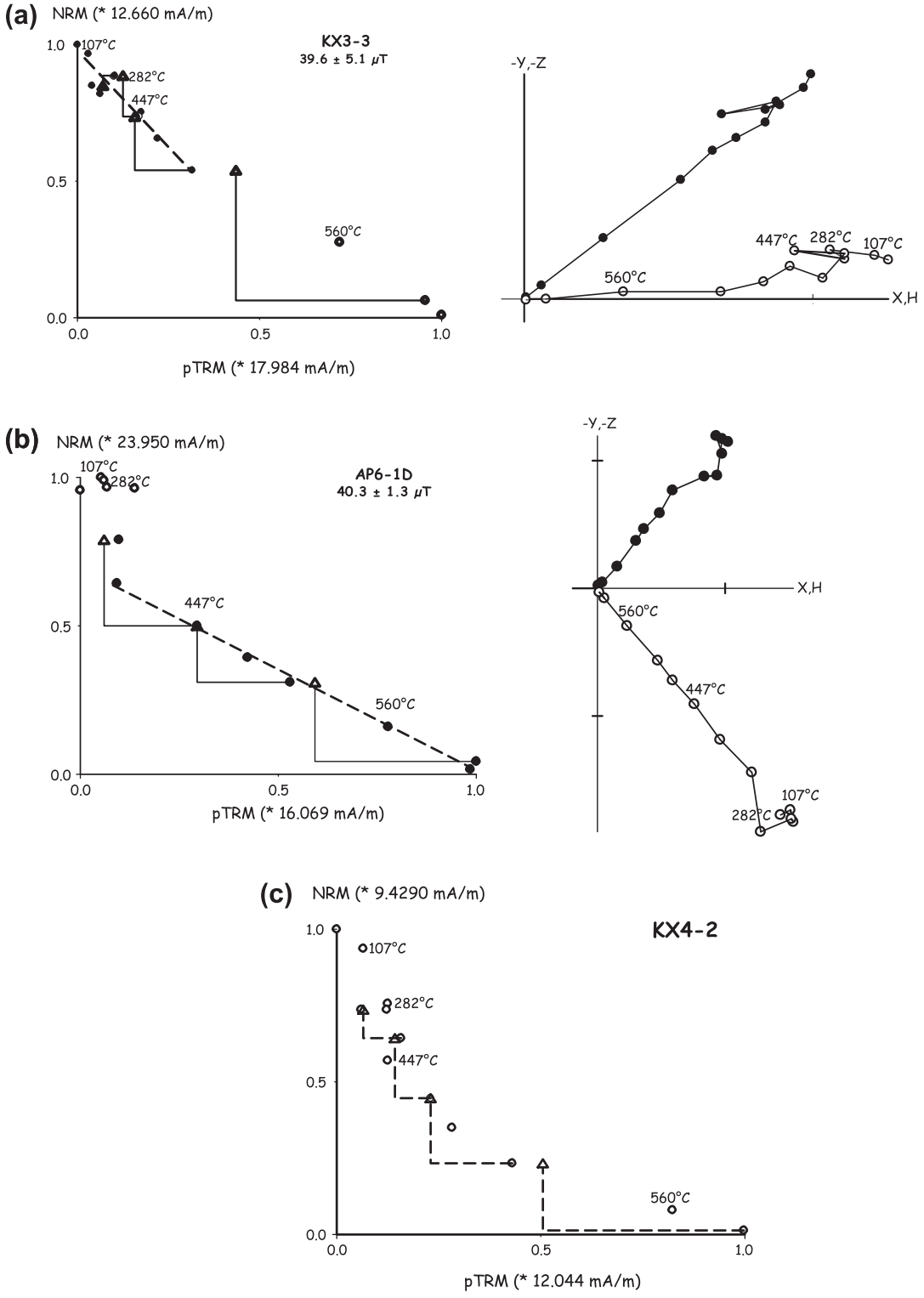


Fig. 7. Paleointensity determinations. (a) NRM–TRM plot of successful determination of specimen KX3-3 (Korxi). (b) NRM–TRM plot of successful determination of specimen AP6-1D (Apnia). Orthogonal vector plot of NRM demagnetisation during the paleointensity experiments are also shown. (b) Unsuccessful paleointensity experiment on specimen KX4-2 (Korxi). Solid circles represent points used for the paleointensity determinations, open circles represent points excluded from paleointensity determinations and triangles represent pTRM-checks.

Table 3
Paleointensity results. *Flow*: Flow name. T_{MIN}–T_{MAX}: Temperature interval used for paleointensity determination. *N*: number of NRM–pTRM points used for paleointensity determination. *f*, *g* and *q*: fraction of extrapolated NRM used, gap factor, and quality factor (Coe et al., 1978) respectively. The fraction factor *f*, the slope of the best fit line the Arai plot, the gap factor *g* and the quality factor *q* were determined for a chosen segment of the Arai diagram, and fraction factor *f* is referred to the so-called "true NRM", which is the intersection between linear fit and *y*-axis (Leonhardt et al., 2004); *sd*/*sl*: Ratio of the standard error of the slope and the slope of the NRM–TRM diagram. *DRAT*: Difference ratio (Selkin and Tauxe, 2000) of PTRM-checks. $d(^*t)$: normalised pTRM-tail $d(t^*)$ (Leonhardt et al., 2004b). *MAD*: Mean angular deviation of NRM end-point directions at each step obtained from paleointensity experiments. $F \pm \Delta F$: Paleointensity estimate for a single specimen and its standard error. $F_M \pm \Delta F_M$: Paleointensity estimate for a site (lava flow) and its standard deviation. *M*: Virtual dipole moment. $M_M \pm \Delta M_M$: Mean virtual dipole moment for a site (lava flow) and its standard deviation.

Flow	T_{MIN}–T_{MAX} (°C)	*N*	*f*	*g*	*q*	*sd*/*sl*	DRAT (°)	$d(t^*)$ (°)	MAD (°)	$F \pm \Delta F$ (μT)	$F_M \pm \Delta F_M$ (μT)	M (10^{22} A m^2)	$M_M \pm \Delta M_M$ (10^{22} A m^2)
AP1-1D	25–525	9	0.80	0.80	6.30	0.10	2.3	2.4	2.3	76.1 ± 7.7	76.1	13.51	13.5
AP2-1D	400–628	5	0.90	0.70	11.9	0.05	3.5	1.3	0.7	26.0 ± 1.4	42 ± 22	4.58	7.4 ± 3.9
AP2-4D	25–525	9	0.35	0.76	2.8	0.10	5.0	1.4	1.4	57.6 ± 5.6		10.14	
AP5-1D	349–525	4	0.51	0.62	4.4	0.07	4.6	4.0	3.4	19.6 ± 1.4	17.3 ± 1.6	3.94	3.5 ± 0.4
AP5-2B	400–629	7	0.88	0.82	24.7	0.03	7.0	2.6	3.0	15.3 ± 0.4		3.08	
AP5-3C	349–629	7	0.88	0.81	13.3	0.05	6.5	3.2	2.5	17.3 ± 0.9		3.49	
AP5-4C	282–629	8	0.98	0.85	13.1	0.06	7.0	2.4	2.6	16.9 ± 1.1		3.41	
AP6-1D	400–629	7	0.89	0.79	21.6	0. 03	4.9	5.7	1.0	40.3 ± 1.3	39.8 ± 7.8	8.11	8.3 ± 0.4
AP6-2D	400–629	7	0.88	0.81	22.6	0.03	6.5	6.2	2.4	42.5 ± 1.4		8.55	
AP6-3A	400–629	7	0.88	0.81	29.8	0.02	4.7	3.1	1.8	43.6 ± 1.0		8.77	
AP6-5A	400–595	6	0.84	0.79	11.4	0.06	6.7	1.4	3.2	42.7 ± 2.5		8.59	
AP6-5C	400–595	6	0.86	0.79	30.2	0.02	3.0	4.9	1.5	37.4 ± 0.9		7.53	
AP6-6C	400–629	7	0.84	0.82	10.8	0.06	3.9	4.9	1.9	41.9 ± 2.7		8.43	
AP7-2B	25–629	12	0.98	0.75	18.1	0.04	2.9	3.8	0.9	54.3 ± 2.2	54.3	9.09	9.1
AP10-1D	400–595	6	0.98	0.72	7.30	0.10	2.6	4.9	2.1	23.9 ± 2.3	35 ± 18	4.33	6.4 ± 2.4
AP10-2D	25–595	11	0.99	0.73	14.9	0.05	5.5	1.5	1.3	31.9 ± 1.6		5.79	
AP10-4D	107–525	8	0.44	0.72	3.6	0.10	1.5	0.5	2.7	49.4 ± 3.8		8.96	
KX2-3	447–629	6	0.86	0.77	4.1	0.08	1.8	2.2	2.8	22.7 ± 1.2	22.7	4.56	4.6
KX3-3	25–525	10	0.46	0.65	2.7	0.11	6.9	3.5	2.6	40.5 ± 4.4	40.5	7.96	8.0

western Djavakheti Highland in southern Georgia. Characteristic remanent magnetisation directions were determined in all 14 lava flows and they are considered to be of primary origin. Normal polarity directions could be clearly recognised in six flows, reversed polarity directions in seven flows and a transitional direction was unmistakably recorded in one flow (site AP2).

Paleomagnetic poles were calculated for both the Apnia and the Korxi sections from VGPs obtained from individual flows (Table 2). In the Apnia section site AP2 was excluded from the calculation. An excellent agreement is observed if the pole obtained from the Apnia flows (latitude $\lambda = 81.8°$N, longitude $\phi = 173.1°$E, $A_{95} = 15.6°$, $k = 11.9$) is compared with the 5 Ma window of the synthetic apparent polar wander path constructed for Europe by Besse and Courtillot (2002) (Fig. 7). If the same comparison is performed with the pole calculated from the Korxi flows (latitude $\lambda = 67.5°$N, longitude $\phi = 135.5°$E, $A_{95} = 18.3°$, $k = 26.9$) a relatively large angle of 19.6° is observed between both poles (Fig. 8).As only four flows from the lowermost part of the Korxi section were analysed in the present study, it is probable that secular variation has not been averaged out. In order to check this in the studied sections, the angular variance of the VGPs of the Apnia, Korxi and the combined Apnia-Korxi datasets – the total angular dispersion $S_T = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}\delta_i^2}$ (Cox, 1969) – was calculated with respect to the pole of the Earth's rotation axis (*N* is the number of sites used in the calculation and δ_i the angular distance of the *i*th VGP from the axial dipole). The optimum cut-off angle was again determined with the method of Vandamme (1994), this time to evaluate ASD with respect to the pole of the Earth's rotation axis, and again only site AP2 had to be excluded from the Apnia data. In addition, it was necessary to correct for the within-site angular dispersion S_W (McElhinny and McFadden, 1997), so that the corrected total angular dispersion S_B is given by $S_B^2 = S_T^2 + \frac{S_W^2}{\bar{n}}$, where $\bar{n}$ is the average number of samples measured in each flow. Upper and lower confidence limits (S_{up} and S_{low}) for the total angular dispersion S_B were calculated following Cox (1969). If results are compared with predicted dispersions calculated from specific models like phenomenological Model G (McFadden et al., 1988), it can be observed that VGPs from the Korxi section show a much higher scatter than expected (Fig. 9). In order to analyse this result it is important to note that while $S_T = 15.4°$ when the ASD for the Korxi section is calculated with respect to the mean of its four VGPs, a value of $S_T = 35.2°$ is obtained when it is calculated relative to the pole of

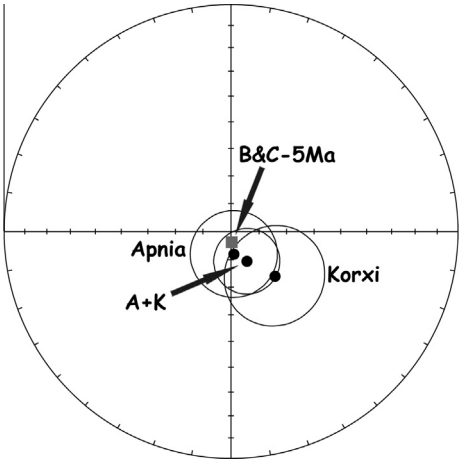


Fig. 8. Stereographic projection of the paleomagnetic poles (solid circles) and 95% confidence cones of the Apnia, Korxi and both combined sequences. Poles were calculated using only flows with normal or reversed polarities). A grey square shows the pole corresponding to the 5 Ma window of the European synthetic polar wander path of Besse and Courtillot (2002).

the Earth's rotation axis. This means that either the Korxi data are disturbed by some tectonic rotation or that they only span an interval insufficient to average out secular variation. The fact that both normal and reversed polarities are observed among the Korxi data points to the first interpretation. On the other hand, if the previous comparison of ASD relative to the VGP mean and the pole of the Earth's rotation axis is performed on the combined Apnia-Korxi dataset, it can be observed that angular dispersion only displays a slight change ($S_T = 22.5°$ relative to the VGP mean and $S_T = 25.5°$ relative to the Earth's rotation axis), which means that the Korxi fit well into that secular variation pattern. It should be nevertheless mentioned that also the VGPs from the Apnia section and the combined Apnia-Korxi VGPs show a higher scatter than expected (Fig. 9). Nevertheless, their lower confidence limits S_{low} appear relatively near to the predicted model G values for the latitude $\lambda = 41.4°N$

No radiometric age data are available for the Korxi section and estimation of its age is only based on a comparison of its petrologic characteristics with those of the Apnia section (Lebedev et al., 2004), but as Pleistocene and Pliocene magmatic activity in the western Djavakheti Highland lasted approximately from 3.75 to 1.55 (Lebedev et al., 2008), data from both sections can be merged in order to provide paleomagnetic information about the upper Pliocene (Piacenzian) to Pleistocene (the latter if a maximum uncertainty for the age of the Korxi section is considered). In addition, if both sets of paleomagnetic directions are compared with an *F*-test in order to find out if the means of both data sets can be discriminated from one another (without considering site AP2 because of its intermediate polarity direction), both mean directions do not appear to be different at a 95% level of probability,

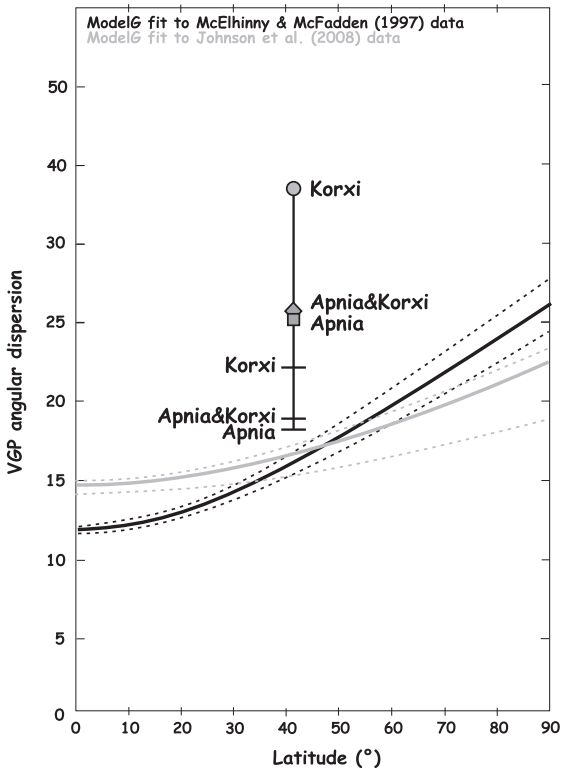


Fig. 9. Virtual Geomagnetic Pole (VGP) angular dispersion versus latitude of paleosecular variation of lavas (PSVLs) for the last 5 Ma shown with Model G fits to data from McElhinny and McFadden (1997) (in black) and Johnson et al. (2008) (in grey). Angular dispersion of VGPs of the Apnia, Korxi and combined Apnia and Korxi sections is shown. For the sake of clarity, only lower confidence limits of the data from the present study are shown.

If the poles from the four Korxi flows are included in a mean Apnia/Korxi pole, a paleomagnetic pole with latitude $\lambda = 77.9°N$ and longitude $\phi = 152.1°E$, ($A_{95} = 11.8°$, $k = 13.4$) is determined, which also shows a good agreement with the 5 Ma window of the synthetic polar wander path for Europe from Besse and Courtillot (2002) (Fig. 8).

The determination of tectonic motion from paleomagnetic data may be accomplished by the comparison of the results obtained for declination and inclination in the studied units (D_0 and I_0) with the expected declination and inclination values (D_{EX} and I_{EX}) calculating rotation $R = D_0 - D_{EX}$ and flattening $F = I_{EX} - I_0$. If a paleomagnetic direction for the Apnia and Korxi sites is determined from the 5 Ma window of the European synthetic APWP (Besse and Courtillot, 2002), an expected direction D = 3.7° and I = 58.3° is obtained. Comparison with the results obtained in both sections after calculation of confidence limits (Demarest, 1983), shows that measured and expected inclinations show an excellent agreement ($F_{APNIA} = 3.3° \pm 9.6°$ and $F_{KORXI} = 3.0° \pm 11.1°$). Declinations, however, match well only in the case of the Apnia section ($R_{APNIA} = 3.9° \pm 17.1°$). The declination difference in the Korxi section amounts to almost 24° ($R_{KORXI} = 23.9° \pm 20.1°$), although the experimental uncertainty $\Delta R = \pm 20.1°$ must be also taken into account. This means that a significant clockwise rotation of the Korxi section cannot completely ruled out, although the result obtained can also be explained with the relatively high experimental uncertainty observed in this section. This high uncertainty is related to the fact that only four flows were available in the Korxi section, resulting in a high value of the α_{95} radius of the confidence cone of its mean paleomagnetic direction. On the other hand, as previously mentioned, not averaged secular variation could be the cause of the rotated paleodeclination observed in the Korxi section.

The mean direction of the 13 sites showing normal and reversed polarities from both the Apnia and the Korxi sections was also calculated (Table 2) and compared with the expected one from the 5 Ma window of the European synthetic APWP (Besse and Courtillot, 2002) and Plio-Quaternary directions obtained in previous studies in Georgia (Goguitchaichvili et al., 2000, 2001; Calvo-Rathert et al., 2011). Declination and inclination of the combined Apnia-Korxi dataset (13 flows) agrees well with the expected ones from the European synthetic APWP ($R_{AP\&KO} = 10.1° \pm 13.3°$; $F_{A\&KO} = 2.4° \pm 7.3°$) and a good agreement was also observed between the paleomagnetic direction obtained in the present study and Plio-Quaternary directions of the previous works, with angular differences between 3° and 5°, which are smaller than the α_{95} cone of the paleomagnetic direction obtained in the present study. This agreement could be observed separately both in inclination and declination after calculation of confidence limits (Demarest, 1983). Analysis of paleomagnetic poles and ChRM directions thus suggests that no significant tectonic rotations have taken place in none of both studied sections. This is in accordance with the fact that the South/North relative motion between the Arabian and Eurasian plates is accommodated in the Southern Caucasus mainly by crustal shortening (Adamia et al., 2008 and references therein). Submeridional compression reaches its peak within the central segment of the Caucasus (Djavakheti-Dzirula salient). (Adamia et al., 2008). NE–SW left-lateral strike-slip faults are main seismoactive structures in the south-west (NE-Turkey) and right-lateral strike-slip faults have developed in the East (south Armenia, NW Iran) (Adamia et al., 2011).

Paleointensity determinations using the Coe (1967) double heating method were carried out on 31 non-oriented samples from 10 lava flows. After taking into account an array of specific selection criteria 19 determinations (61%) appeared to provide successful paleointensity results (Table 3), which can be considered a rather high success rate. Positive results, however, were obtained on specimens from eight flows, specifically in 17 out of 27

specimens from the Apnia sequence and 2 out of 4 in the Korxi sequence. In addition, only one successful determination was obtained from flows AP1, AP7, KX2 and KX3, as in these sites only a single specimen was available for paleointensity determinations. Most determinations were performed on samples from flows AP2 (two out of five successful determinations), AP5 (four out of five successful determinations), AP6 (6 out of 10 successful determinations) and AP10 (three out of four successful determinations). Flows AP2 and AP5, as well as AP1, AP7 and KX3 are characterised by type-H thermomagnetic curves, but in flows AP6, AP10 and KX2 type H^- curves were observed (Table 1). Although in both curve types magnetite is the only ferromagnetic (s.l.) phase observed in both heating and cooling curves, type H curves are fully reversible while in type H^- curves an intensity decrease is observed in the cooling curve. This means, that either magneto-mineralogical transformations responsible for the irreversibility observed in type H-curves occur at rather high temperatures or different thermomagnetic behaviours do occur in nearby lying samples from the same flow. If domain size characteristics are considered, flows AP1, AP6, AP7 and KX2 show the most SD-like characteristics (Figs. 4a and 5). Among these sites, only in flow AP6 several successful determinations are available, and a low scatter of paleointensity results is observed.

The mean paleointensity values of the studied flows lie in wide range, between 17 and 76 μT (Table 3). However, as already mentioned, in four flows only a single paleointensity determination is available, but on the other hand, several successful paleointesity determinations could be obtained in sites AP2, AP5, AP6 and AP10. For comparison, Goguitchaichvili et al. (2009) found a Pliocene field $B_{PLIOCENE}$ = 34 μT in the western Djavakheti area and Calvo-Rathert et al. (2011) obtained Pliocene and Pleistocene paleointensity values mainly in the 30–45 μT range in the same study region. The present day field in Georgia has an approximate value $B_{PRESENT}$ = 47 μT. Paleointensities obtained from transitionally magnetised flow AP2 deserve special attention. No weak transitional paleostrength values were observed, although the two results obtained differ significantly. On the other hand, reversed polarity flow AP5 displays a rather weak paleointensity based on four successful paleointensity determinations of similar strength. It should be borne in mind that the flows analysed in the present study belong to two lava flow sequences, Apnia and Korxi, which have a significantly larger number of flows than those sampled for the study. In addition, the position of each of the studied flows within the sequence to which it belongs is unknown. As the magnetisation carried by volcanic rocks is only able to provide an instant view of the behaviour of the Earth's magnetic field, a deeper interpretation of the paleointensity variation among different flows would need the study of the whole Apnia and Korxi sites with knowledge of the position of the flows within both sequences.

Table 3 shows virtual dipole moment (VDM) values calculated for all 19 paleointesities obtained in the present study. For the calculation, mean site inclinations have been taken. Fig. 10 shows four mean site VDMs from the present study after excluding those from sites in which only a single paleointensity determination was available. Three of these flows (AP5, AP6 and AP10) display values between 6.4×10^{22} A m^2 and 8.3×10^{22} A m^2 while the remaining one shows a rather weak VDM of 3.5×10^{22} A m^2. As can be recognised in Fig. 10, all four results fit into the rather scattered range of Pliocene VDM values (2–5 Ma) obtained by means of Thellier-type and microwave paleointensity experiments performed with pTRM-checks (PINT database, Biggin et al., 2010). Data points from this database shown in Fig. 10 do not include VDMs from transitional magnetisation vectors, although results from the present study apparently corresponding to intermediate polarities (AP2) have been included in the figure, as they are characterised by paleointensities not differing from the expected ones in Georgia during non-transitional periods in the Pliocene.

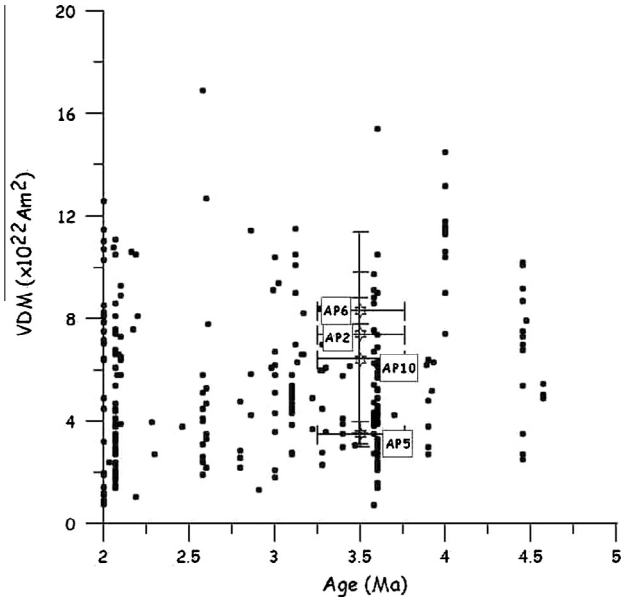


Fig. 10. Pliocene (2–5 Ma) VDM obtained from Thellier-type and microwave paleointensity experiments with pTRM-checks. Data have been taken from the PINT paleointensity database (Biggin et al., 2010) and data from transitional magnetisation vectors have been omitted. White stars show results from the present study with error bars.

7. Main results and conclusions

Paleomagnetic, rock-magnetic and paleointensity measurements have been carried out on 14 basaltic lava flows of Pliocene age (K–Ar age between 3.09 ± 0.10 Ma and 4.00 ± 0.15 Ma) from the eastern Djavakheti Highland in southern Georgia. These lava flows belong to two different lava flow sequences, Apnia and Korxi. These studies were directed towards two aims: (i) Obtaining information to contribute to the description of the variations of the intensity and direction of Earth's magnetic field and (ii) supplying new paleomagnetic data from the Caucasus and specifically from the Djavakheti Highland which might be of interest to obtain more information about the recent tectonics in the study region.

Characteristic remanent magnetisation directions of primary origin could be determined in all 14 lava flows. Normal polarity directions were recognised in six flows, reversed polarity directions in seven flows and a transitional direction was recorded in one flow (Table 2). While the 5 Ma window of the European synthetic polar wander path (Besse and Courtillot, 2002) shows an excellent agreement with the mean pole obtained from nine flows from the Apnia section, it forms an angle of 19.6° with the pole calculated from the Korxi flows (Fig. 8). The reason of this discrepancy could be nonaveraged secular variation or a tectonic rotation of the Korxi section. While no significant declination difference was observed between the mean paleomagnetic direction of the Apnia section and the expected direction, a rotation of 23.9° ± 20.1° was recognised between the latter and the Korxi section. A combined pole obtained from 13 lava flows from both the Apnia and the Korxi sections agrees, however, with the 5 Ma window of the synthetic polar wander path for Europe and shows no significant of paleodeclination (Fig. 8). This absence of rotation would be in accordance with the fact that the South/North relative motion between the Arabian and Eurasian plates is accommodated in the

Southern Caucasus mainly by submeridional crustal shortening, in the direction of plate convergence.

Analysis of the angular dispersion of VGPs of the Korxi section yields a much higher value than expected by phenomenological Model G (Fig. 9) (McFadden et al., 1988). Also the VGPs from the Apnia section and the combined Apnia-Korxi VGPs display a higher scatter than expected, but their lower confidence limit S_{low} appear relatively near to the predicted model G values for the latitude $\lambda = 41.4°N$ (Fig. 9).

Paleointensity experiments were performed with the Coe method on 31 specimens from 10 of the 14 studied flows. After application of specific selection criteria, 19 samples from 8 flows were observed to provide successful determinations. The mean paleointensity values of the studied flows lie in a wide range, between 17 and 76 μT (Table 3). For comparison, Pliocene and Pleistocene paleointensity values found in recent studies in the Djavakheti region (Goguitchaichvili et al., 2009; Calvo-Rathert et al., 2011) lie in the 30–45 μT range. If only flows with more than one successful paleointensity determination are taken into account, virtual dipole moments (VDMs) vary between 3.5×10^{22} A m^2 and 8.3×10^{22} A m^2 (Fig. 10). In site AP2, which records an intermediate polarity direction, no weak transitional paleostrength values was observed.

A future study of the whole lava flow sequences of both the Apnia and Korxi sections would be of great interest in order to complete the paleomagnetic and paleointensity results obtained in the present study and clarify ambiguities in the interpretation of the Korxi data.

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